

The background features a light blue gradient with abstract circuit-like patterns. Purple and orange lines, some with circular nodes, crisscross the frame. In the bottom right, there's a grid of small blue dots with a cluster of larger, semi-transparent blue cubes. The title 'SQIsign' is prominently displayed in the center-left.

SQIsign

By: Sadia Sharmin, Zhian Maysoon, Farha Ferdous
and Najee Mozie

Meet the Team:



Sadia Sharmin



Zhian Maysoon



Farha Ferdous



Najee Mozie

Motivation

- Quantum computers can solve once infeasible tasks
- Need for new cryptographic systems that can withstand quantum attacks
- SQLsign is a strong candidate

A black silhouette of a skier in mid-jump, positioned as if leaping over the word "SQLsign". The skier is holding poles and has their skis angled upwards. The word "SQLsign" is written in a large, elegant, cursive script. The "SQL" part is in a bold, sans-serif font, while "sign" is in a flowing cursive. The skier's jump is positioned directly over the "sign" portion of the word.

SQLsign

Contributors

- Five original developers: Luca De Feo, David Kohel, Antonin Leroux, Christophe Petit, and Benjamin Wesolowski
- Latest submission had 30 authors listed

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SQIsign Overview

- Short Quaternion and Isogeny Signature
- Isogeny-based system
 - Security based on two computationally hard problems:
 - Finding isogenies between supersingular elliptic curves
 - Determining their endomorphism rings
- Key generation → Signing → Verification

SQIsign Overview

- Advantages:
 - Compact keys and signatures
 - Efficient
 - Believed to be resistant to quantum attacks
- Limitations:
 - Complex signing procedure

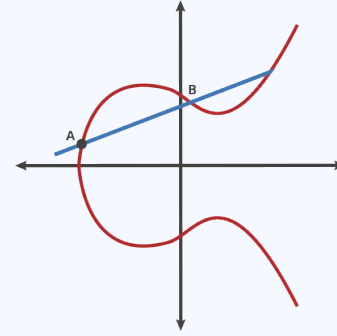
TABLE 1. SQISIGN key and signature sizes in bytes for each security level.

Parameter set	Public key	Secret key	Signature
NIST-I	65	353	148
NIST-III	97	529	224
NIST-V	129	701	292

Background

Background & Preliminaries for SQLsign

To understand SQLsign, one must first be familiar with the core mathematical constructs it is built upon. SQLsign's security relies on complex mathematical constructs that are believed to be resistant to quantum attacks. The following will provide a detailed overview of these foundations, focusing on the concepts that are critical to understanding the scheme's internal operations, particularly the Deuring correspondence and the algorithms that leverage it.



Agenda & The Core Challenge

What We Need to Understand

- Elliptic Curves and Isogenies
- Supersingular Curves
- Why Quantum Computers Get Stuck
- The Deuring Correspondence
- The Fiat-Shamir Transformation

Concept 1: Elliptic Curves & Isogenies

Curves and Bridges

- **Elliptic Curve:** A set of points forming an abelian group. The foundation of ECC.
- **Isogeny:** A special map (a "bridge") between two elliptic curves that preserves the group structure.
- **Key Insight:** You can create a vast graph where each vertex is a curve and each edge is an isogeny. This is the isogeny graph.

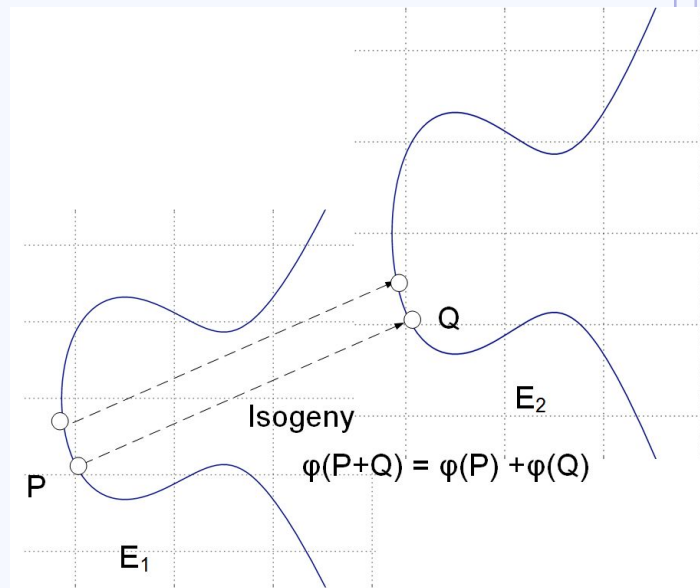
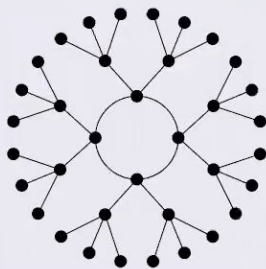


Image from Medium article: *Goodbye to Elliptic Curves? ... Not, Quite ... Here Come Isogenies*

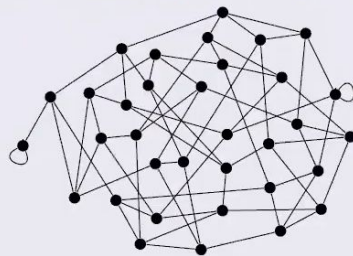
Concept 2: Supersingular Elliptic Curves

- **Ordinary Curves:** Have a commutative endomorphism ring. Like a simple, predictable grid map.
- **Supersingular Curves:** Have a non-commutative endomorphism ring. The "magical" islands where the geography is chaotic.
- **Why it matters:** This non-commutativity means the order of operations matters. This chaos is what makes creating a map incredibly hard for an attacker.

An ordinary component:



The supersingular component:



Images from YouTube video:
*Supersingular curves with small
non-integer endomorphisms* by ANTS

Concept 3: The Hard Problems

- **Supersingular Isogeny (SSI) Problem:** "Given two supersingular curves, find the isogeny (bridge) connecting them." This is like finding a single, specific path in a galaxy of stars.
- **Endomorphism Ring Problem:** "For a given supersingular curve, figure out its entire, chaotic road system." Knowing this is equivalent to having a master key.
- **Quantum Resistance:** Shor's algorithm breaks RSA/ECC by exploiting their commutative structure. It has no power over this non-commutative chaos.

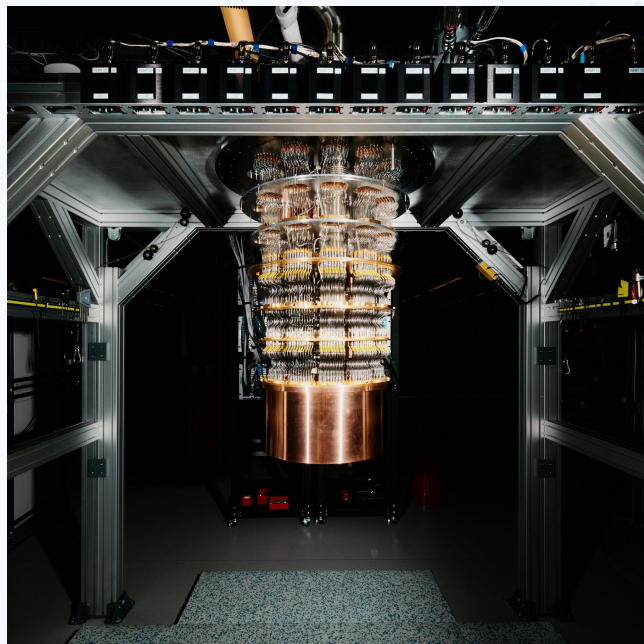


Image from NYT

Concept 4: The Deuring Correspondence

- A fundamental mathematical **bijection** (a perfect pairing).
- It connects the world of **Supersingular Elliptic Curves** to the world of **Quaternion Algebras**.
- **Why SQLsign needs it:** Some calculations are incredibly hard in the "curve world" but become manageable in the "quaternion world." Deuring lets us translate back and forth.

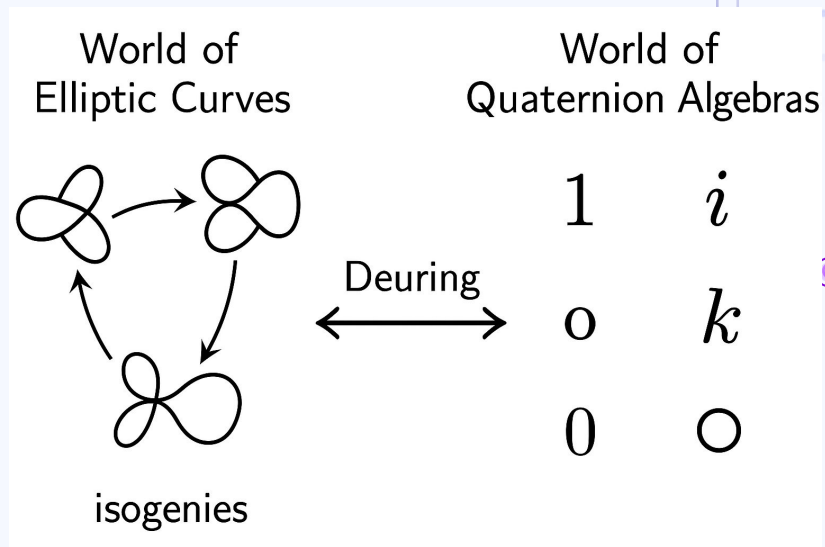


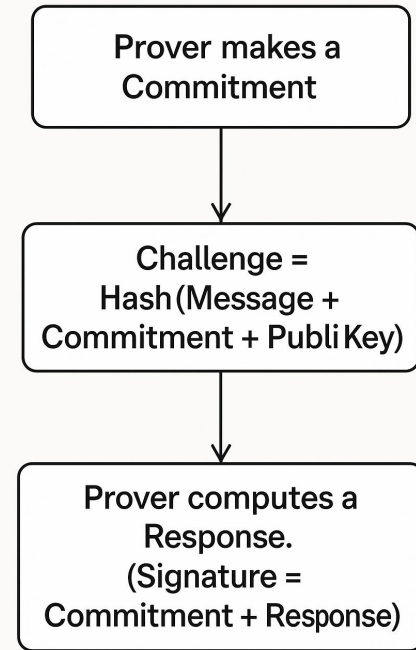
Image generated using ChatGPT

Concept 5: Fiat-Shamir Transform

- SQLsign starts as an interactive proof: "I know a secret" -> "Prove it" -> "Here's proof."
- **Fiat-Shamir** removes the verifier! It uses a cryptographic hash function to generate the challenge.
- This converts the interactive proof into a **non-interactive digital signature**.

What This Means:

- Removes the need for a live "verifier" to ask a random question.
- The challenge is now deterministic and unique to the message.
- This creates a standalone proof that anyone can verify.
- SQL needs this because it converts the interactive identification protocol into a practical digital signature you can send in an email or attach to a document.



Summary: Tying It All Together

- We now have a quantum-resistant foundation based on hard math problems.
- We have an efficient engine **(Deuring)** for compact keys and signatures.
- We have a method to create non-interactive signatures **(Fiat-Shamir)**.

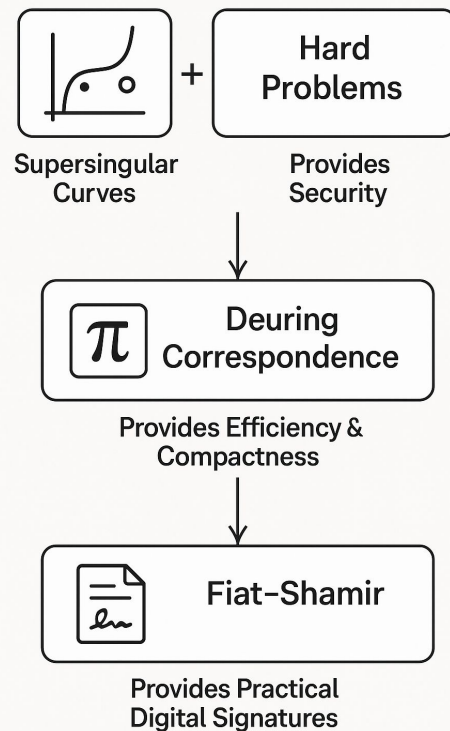


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Description of Scheme



Core Ideas

Cryptographic Keys $\rightarrow$ elliptic curves

Encode Private keys (isogenies) $\rightarrow$ to be the secret path

Public key is an end point of the path (of the curve that can be publicly known)

Core Ideas

- **Private Key:** A secret "path" (an **isogeny**, ϕ) between two elliptic curves.
- **Public Key:** The ending curve (E_A) of that path.
- **Security:** Finding the path between the public starting curve and the public ending curve is computationally impossible for an attacker.

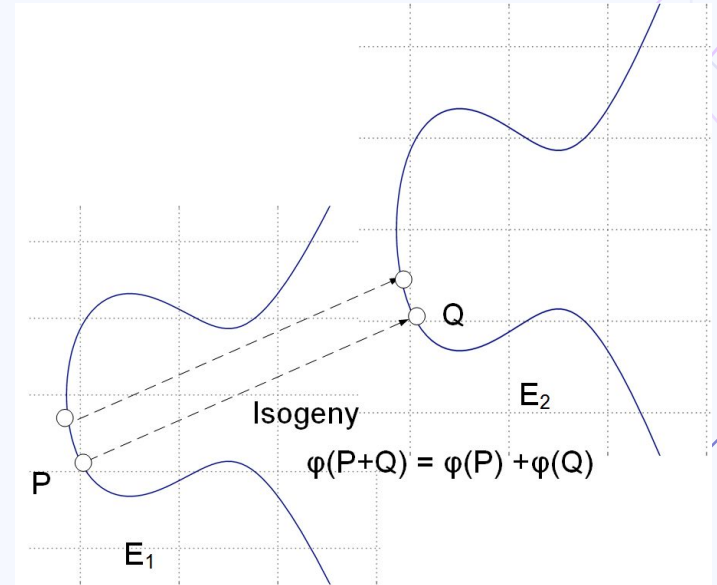


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The 3 core phases of SQI Sign

1. **Key Generation:** Creating the secret path and the public destination.
2. **Signing:** Using the secret path to create a proof for a specific message.
3. **Verification:** Allowing anyone to check the proof without knowing the path.

Core Phase: Key Generation

Input:

- Public starting curve E_0

Process:

- Signer randomly chooses a secret isogeny, φ .

Output:

- **Private Key:** The isogeny φ .
- **Public Key:** The resulting curve E_a .

Core Phase: Signing

1. **Commitment:** Prover creates a temporary curve E_1 and sends it.
2. **Challenge:** Verifier sends a random challenge.
3. **Response:** Prover uses their *secret* φ to compute a response σ .

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But how do we standardize this? We can't have live dialogue the whole time.

Core Phase: Signing

We can use the non-interactive zero knowledge proof, based on the Fiat-Shamir transformation. This allows proof to remain secure while eliminating the need for interaction between parties (Signer and Verifier).

But what is the Fiat-Shamir Transformation?

Zero Knowledge Proof

Protocol between two parties, a **Prover (P)** and a **Verifier (V)**, which allows **P** to convince **V** that they know a certain statement

For a protocol to be a zero-knowledge proof, it must satisfy the following three properties:

1. **Completeness:** If the prover is honest and knows the secret, then an honest verifier will be convinced of this fact. In other words, a valid proof always succeeds.
2. **Soundness:** If the prover does *not* know the secret, then they cannot convince a verifier that they do, except with some negligibly small probability. A cheating prover will almost always be caught.
3. **Zero-Knowledge:** The verifier learns *nothing* from the interaction other than the fact that the statement is true.

Core Phase: Verification

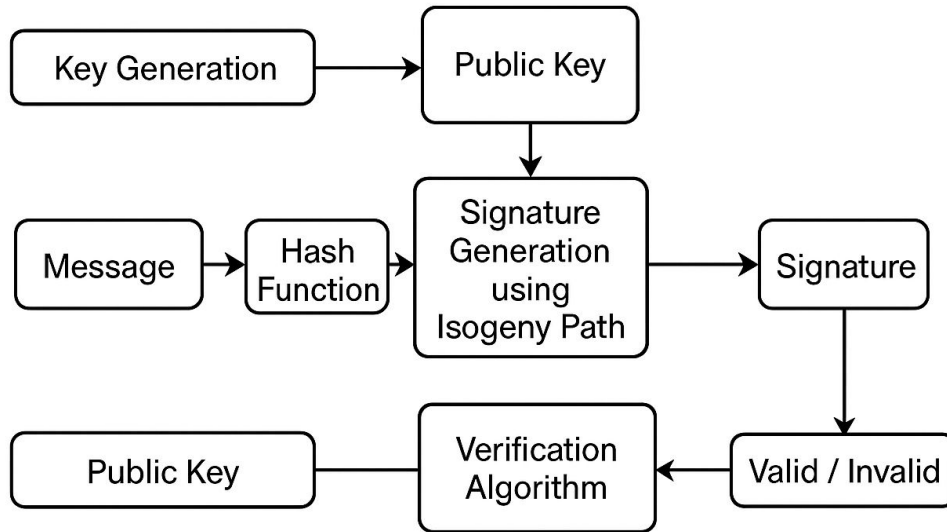
What the Verifier Has:

- Message m , Signature (E_1, σ) , Public Key E_a .

What the Verifier Does:

1. Recomputes the challenge: $\text{Hash}(m \parallel E_1)$.
2. Uses the public key E_a and the response σ to walk through a cycle of curves.
3. **The Key Check:** Verifies that this cycle completes and returns to the starting point, E_0 .

Diagrams



Algorithm Overview

- The SQIsign algorithm produces:
 - Key Generation
 - Digital Signatures Creation
 - Verification
- It relies on advanced math based on isogenies between a special kind of elliptic curves.
- This keeps public keys small, only about 204 bytes, while maintaining strong post-quantum security.

Role of Algorithms in SQIsign

- SQIsign is based on the hardness of the Supersingular Isogeny Problem.
- This algorithm defines how SQIsign:
 - Generate unique public and private keys
 - Creates secure digital signatures
 - Verifies authenticity and integrity of messages

Efficient Key Sizes in SQIsign

Scheme	Type	Public Key Size
SQIsign	Isogeny-based	~204 bytes
Dilithium	Lattice-based	~1,312 bytes
Falcon	Lattice-based	~897 bytes
SPHINCS+	Hash-based	~16,000 bytes
RSA-2048 (<i>classical</i>)	Integer factorization	~256 bytes (not quantum secure)

Offers one of the smallest public key size among other post-quantum signature schemes.

Key Generation

- Start from base elliptic curve E_0 over finite field $\mathbb{F}_{q^2}$.
- Choose secret isogeny $\phi: E_0 \rightarrow E_a$ (private key).
- Compute public curve $E_a = \phi(E_0)$.
- Map reference points $P^B, Q^B \rightarrow \phi(P^B), \phi(Q^B)$.

Public Key = $(E_a, \phi(P^B), \phi(Q^B))$

Private Key = ϕ

E_0 is the shared base curve used by all users in the system.

Signing

- Hash the message $m \rightarrow H(m)$
- Use secret ϕ to generate a chain of small-degree isogenies:
$$\mathbf{E}_a \rightarrow \mathbf{E}_1' \rightarrow \mathbf{E}_2' \rightarrow \dots \rightarrow \mathbf{E}_{\square}$$
- The chain acts as a unique signature path

Signature = $\{\mathbf{E}_a, \text{sequence of isogenies} \rightarrow \mathbf{E}_{\square}\}$

Verification

The receiver checks to make sure the message wasn't tampered with and that it actually came from the sender, this is done by:

1. **Recompute hash: $H(m)$**
2. **Rebuild expected curve E' using public key and $H(m)$**
3. **Extract E from signature**
4. **Compare E' and E :**
If $E' = E \rightarrow$ Valid signature

Verification Pseudocode

From

input: message m , signature s , public key $(E_a, \phi(P^B), \phi(Q^B))$

$H_m = \text{Hash}(m)$

$E'_\square = \text{RebuildCurveFromPublicData}(E_a, \phi(P^B), \phi(Q^B), H_m)$

$E_\square = \text{ExtractCurveFromSignature}(s)$

if $E'_\square == E_\square$:

return 'VALID'

else:

return 'INVALID'

Algorithm Summary

1. Key Generation:

Create a private key (**using secret isogeny**) then use it to build a public key (**from the known elliptic curve**).

2. Signing:

Use the private key to create a unique signature path for the message.

3. Verification:

The receiver makes sure the message wasn't tampered with and that it came from the correct person.

The background features a light blue gradient with abstract circuit-like patterns. Purple and orange lines, some with circular nodes, crisscross the frame. In the bottom right, there's a cluster of blue dots and a small 3D cube structure.

SQIsign

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Meet the Team:



Sadia Sharmin



Zhian Maysoon



Farha Ferdous



Najee Mozie



Security Analysis

Advantage

One of the main attractions for SQIsign is its small key sizes

TABLE 1. SQIsign key and signature sizes in bytes for each security level.

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Security

Positives

- No efficient attacks exist
- Believed to be resistant to quantum attacks

Negatives

- Too hard :(
- Complexity makes it difficult to conduct thorough cryptanalysis
- Less confidence in long term stability

Threats

- Primary threats are side-channel attacks
 - Timing attacks
 - The attacker learns the secret key by precisely measuring how long a cryptographic operation takes to complete
 - Power analysis
 - The attacker learns the secret key by monitoring the electrical power consumption of a device while it performs the cryptographic operation
 - Fault injection
 - The attacker deliberately induces a fault or error in the device's operation and then analyzes the incorrect output to deduce the secret key
- Complex code increases the risk of subtle bugs
 - Very few people actually understand the problem



Performance and Parameters

SQIsign Parameters: What They Are & Why They Matter

- **Prime p** defining the field $\mathbb{F}_{p^2}$
- Security levels (NIST-I, III, V)
- **Fiat-Shamir** hash (SHAKE-256)
- Isogeny degrees: **large + small**
- All performance depends on these choices

Why the Prime p Is Chosen Carefully

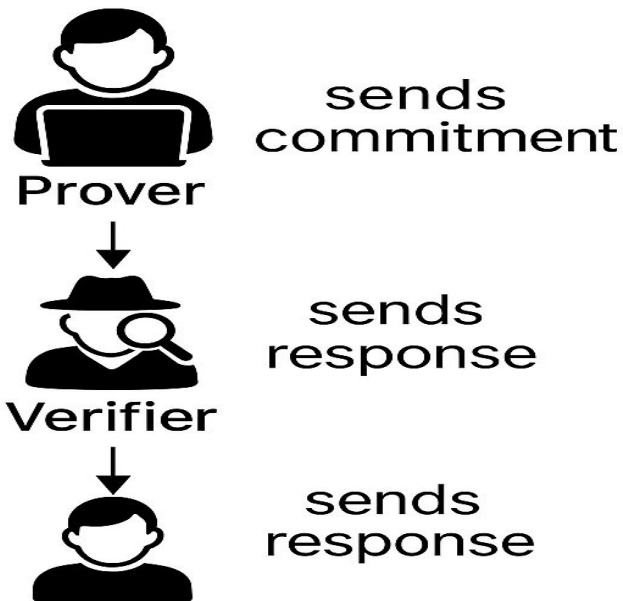
- p selected so $p + 1$ breaks into small primes for fast isogenies
- Very large $\rightarrow$ quantum resistance
- Supports efficient quaternion arithmetic
- Matches curves via **Deuring's correspondence**
- Determines key sizes

Two Isogenies: The Heart of SQIsign

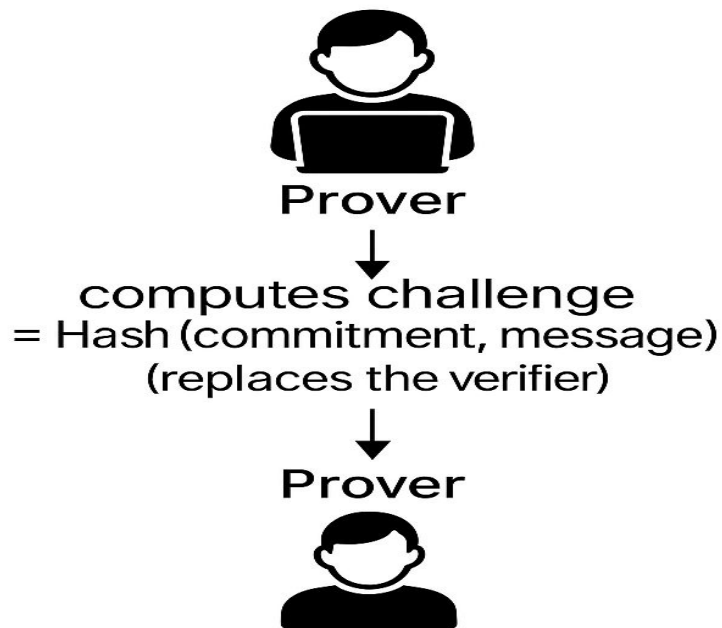
- Large-degree isogeny = **private key**
- Small-degree isogenies = fast steps
- Balance $\rightarrow$ security & speed
- Affects signature size & signing cost

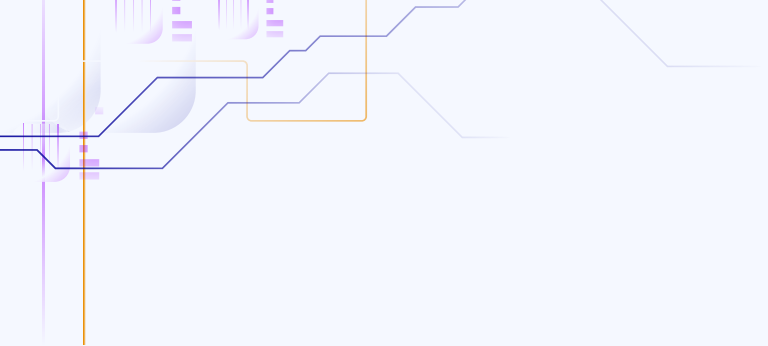
How Fiat-Shamir Impacts Performance

Interactive Authentication



Fiat-Shamir





Comparison

Not All Post-Quantum Schemes Are the Same

- **Lattice-based (e.g., Dilithium):** Relies on the hardness of lattice problems. A leading NIST finalist.
- **Hash-based (e.g., SPHINCS+):** Relies on the security of hash functions. Very conservative security.
- **Isogeny-based (SQIsign):** Relies on the hardness of finding paths between elliptic curves.

Signature Size

RSA-2048: ~256 bytes

Dilithium: ~2,000+ bytes

SPHINCS+: ~8,000+ bytes

SQISIGN: ~177 bytes

The Great Trade-Off: Size vs. Speed

Comparison:

- **SQIsign: Very Slow** Signing (seconds), Fast Verification.
- **Dilithium: Fast** Signing and Verification.
- **SPHINCS+: Slow** Signing and Verification.

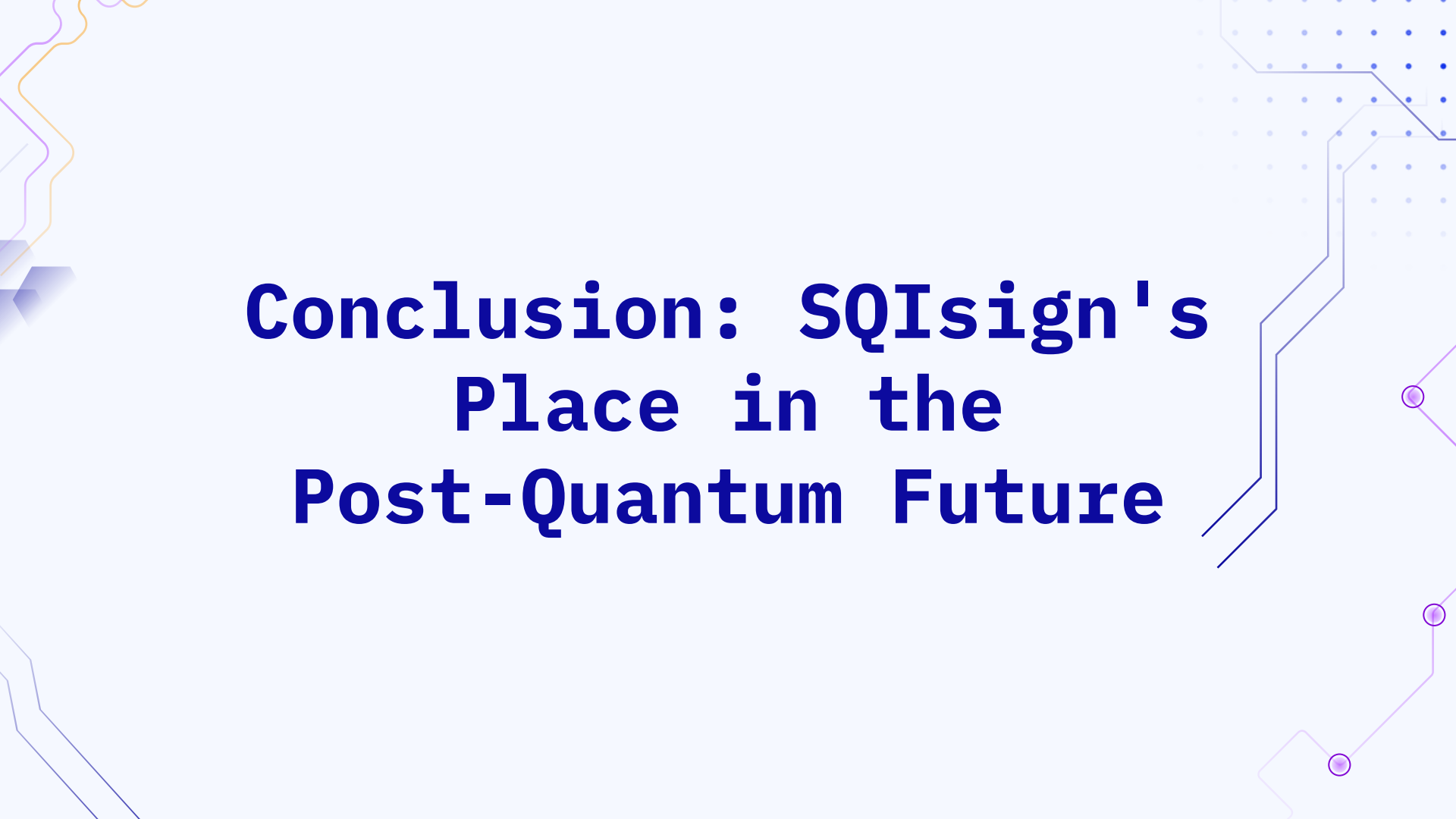
The Maturity & Confidence Factor

Dilithium (Lattice-based):

- **Extensively studied** for over a decade.
- NIST's **primary chosen candidate** for standardization.
- Seen as the **safe, conservative choice**.

SQIsign (Isogeny-based):

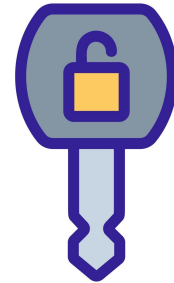
- Relatively **new construction**.
- Still undergoing intensive **cryptanalysis**.
- Higher "confidence risk" but also higher potential reward.

The slide features decorative geometric patterns in the corners. The top-left corner has overlapping wavy lines in purple and orange. The top-right corner has a grid of small blue dots. The bottom-right corner has a purple circuit-like line with circular nodes. The bottom-left corner has a simple purple line.

Conclusion: SQIsign's Place in the Post-Quantum Future

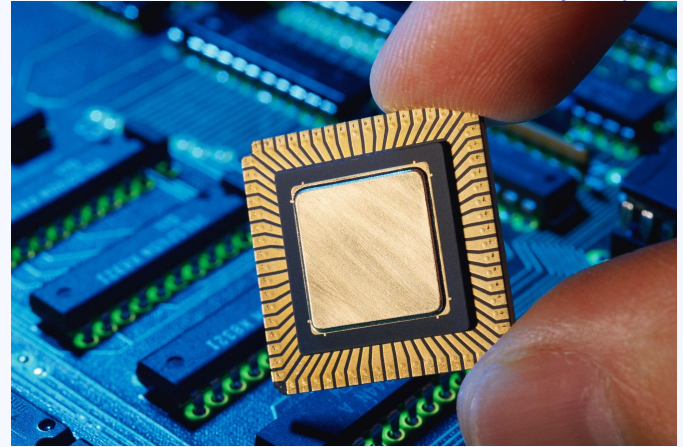
Quantum-Resistant Security

- **Core Hard Problems:** Security rests on the Supersingular Isogeny Problem and the Endomorphism Ring Problem.
- **Mathematical Mismatch:** These problems are based on non-commutative structures, which lack the hidden periodic patterns that Shor's algorithm exploits.
- **Future-Proof:** No known efficient algorithm (classical or quantum) exists to solve them, providing confidence in long-term security.



Unmatched Compactness

- **Minimal Footprint:** Public keys (~64 bytes), Signatures (~148 bytes for NIST-1).
- **Order-of-Magnitude Advantage:** Significantly smaller than lattice-based (kilobytes) and hash-based (tens of kilobytes) proposals.
- **Enables New Applications:** This compactness is a game-changer for IoT sensors, blockchain transactions (lower fees, faster verification), and secure smart cards where memory and bandwidth are severely limited.



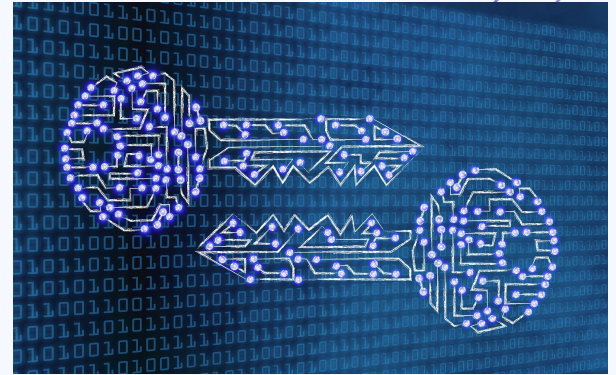
Key Mathematical Innovations

- **The Efficiency Engine:** The Deuring Correspondence + KLPT Algorithm translates hard curve problems into solvable quaternion problems and back, enabling practical computation.
- **The Practicality Bridge:** The Fiat-Shamir Transform converts a complex interactive zero-knowledge proof into a simple, non-interactive digital signature.
- **Theory to Practice:** These two innovations are what transform deep algebraic geometry into a working cryptographic tool.

$$\begin{array}{ccc} E_3 & \xrightarrow{[\psi] * \varphi} & E_4 \\ \psi \uparrow & & \uparrow [\varphi] * \psi \\ E_1 & \xrightarrow{\varphi} & E_2 \end{array}$$

Strengthening the Crypto Ecosystem

- **Mitigates Systemic Risk:** Over-reliance on one mathematical family (e.g., lattices) creates a single point of failure for global security.
- **Provides Independent Security:** SQIsign is built on a completely different foundation, offering an independent line of defense.
- **Strategic Importance:** Cryptographic diversity is a strategic imperative. SQIsign ensures our security infrastructure is resilient, not fragile.



References

Primary Sources:

1. SQIsign Team. (2023). SQIsign Specification (Round 2). NIST PQC Project
2. De Feo, L., et al. (2023). "SQIsign: Compact Post-Quantum Signatures from Quaternion Isogenies." CRYPTO 2023

Additional Resources:

3. National Institute of Standards and Technology. Post-Quantum Cryptography Project
4. "Supersingular Isogeny Key Exchange." Wikipedia